Exceptions

Division par 0

expression $e ::=$ ...
e_1 / e_2 (division)

$$\frac{\Gamma \vdash e_1 : \text{num} \qquad \Gamma \vdash e_2 : \text{num}}{\Gamma \vdash e_1 / e_2 : \text{num}}\ \text{div}$$

Problème : $e_1/0$

Division par 0

expression $e ::=$ ...
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Problème : $e_1/0$

- Vérification statique (système de type) : trop strict
- Vérification dynamique : à l'exécution

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e_1 / e_2 (division)

$$\frac{\Gamma \vdash e_1 : \text{num} \qquad \Gamma \vdash e_2 : \text{num}}{\Gamma \vdash e_1 / e_2 : \text{num}}\ \text{div}$$

Problème : $e_1/0$

- Vérification statique (système de type) : trop strict
- Vérification dynamique : à l'exécution
  - état d'erreur `error`
  - avec ou sans information (entier, chaîne ...)
  - avec ou sans gestion `try ... with ...`
  - personnalisable ou non

λ-calcul avec exceptions

Type

$\tau ::=$ str (type des chaînes)
$\tau_1 \to \tau_2$ (type des fonctions)
type argumt — tyoe résultat

Expression

$e ::=$ x (variable)
s (chaîne littérale)
$e_1 \hat{} e_2$ (concaténation)
$e_1\ e_2$ (application)
`fun` $(x{:}\tau) \Rightarrow e$ (abstraction)
`raise` e (déclencheur)
`try` e_1 `with` $x.e_2$ (récupération)

λ-calcul avec exceptions

Expression

$e ::=$ ...
`raise` e (déclencheur)
`try` e_1 `with` $x.e_2$ (récupération)

Principe :

- `raise "Segmentation fault"`
- `try "coucou" with` $x.e_2 \to$ `"coucou"`
- `try (raise` s`) with` $x.e_2 \to e_2\{s/x\}$

• La première expression signale une exception "Segmentation fault".
• La deuxième expression tente d'évaluer "coucou" sans exception.
• La troisième expression déclenche délibérément une exception et gère la récupération de cette exception avec la clause with

Système de types

Str $\dfrac{}{\Gamma \vdash s : \mathrm{str}}$ Var $\dfrac{}{\Gamma, x:\tau \vdash x:\tau}$ Conc $\dfrac{\Gamma \vdash e_1 : \mathrm{str} \qquad \Gamma \vdash e_2 : \mathrm{str}}{\Gamma \vdash e_1 \hat{} e_2 : \mathrm{str}}$

Fun $\dfrac{\Gamma, x:\tau_1 \vdash e : \tau_2}{\Gamma \vdash \mathtt{fun}\ (x{:}\tau_1) \Rightarrow e : \tau_1 \to \tau_2}$ App $\dfrac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \qquad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1\ e_2 : \tau_2}$

Système de types

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Try $\dfrac{\Gamma \vdash e_1 : ? \qquad \Gamma, x : ?? \vdash e_2 : ???}{\Gamma \vdash \mathtt{try}\ e_1\ \mathtt{with}\ x.e_2 : \tau}$

- $\mathtt{try}\ \text{"coucou"}\ \mathtt{with}\ x.e_2 \to \text{"coucou"}$
- $\mathtt{try}\ (\mathtt{raise}\ s)\ \mathtt{with}\ x.e_2 \to e_2\{s/x\}$

Système de types

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Try $\dfrac{\Gamma \vdash e_1 : \tau \qquad \Gamma, x : \mathrm{str} \vdash e_2 : \tau}{\Gamma \vdash \mathtt{try}\ e_1\ \mathtt{with}\ x.e_2 : \tau}$

- $\mathtt{try}\ \text{"coucou"}\ \mathtt{with}\ x.e_2 \to \text{"coucou"}$ aucune except° levée pdt cette évaluat° résultat est "coucou"
- $\mathtt{try}\ (\mathtt{raise}\ s)\ \mathtt{with}\ x.e_2 \to e_2\{s/x\}$

évalue expression raise s
déclenche except°
en cas d'except°, var x liée à la valeur de l'except°
et e2 évaluée avec sub de x par s

Système de types

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- $\mathtt{try}\ \text{"coucou"}\ \mathtt{with}\ x.e_2 \to \text{"coucou"}$
- $\mathtt{try}\ (\mathtt{raise}\ s)\ \mathtt{with}\ x.e_2 \to e_2\{s/x\}$
- $\mathtt{raise}\ \text{"Segmentation fault"}$

Système de types

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Texte

Fun $\dfrac{\Gamma, x:\tau_1 \vdash e : \tau_2}{\Gamma \vdash \texttt{fun}\ (x{:}\tau_1) \Rightarrow e : \tau_1 \to \tau_2}$ App $\dfrac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \qquad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1\ e_2 : \tau_2}$

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- `try "coucou" with` $x.e_2 \to$ `"coucou"`
- `try (raise` s`) with` $x.e_2 \to e_2\{s/x\}$
- `raise "Segmentation fault"`

Système de types

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- `try "coucou" with` $x.e_2 \to$ `"coucou"`
- `try (raise` s`) with` $x.e_2 \to e_2\{s/x\}$
- `raise "Segmentation fault"`
- `(fun (`$x{:}\tau$`) ⇒` x`) raise "Segmentation fault"`

Système de types

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Dans le contexte Γ, si nous ajoutons l'hypothèse que x a le type τ, alors nous pouvons déduire que x a effectivement le type τ.

Fun $\dfrac{\Gamma, x:\tau_1 \vdash e : \tau_2}{\Gamma \vdash \texttt{fun}\ (x{:}\tau_1) \Rightarrow e : \tau_1 \to \tau_2}$ App $\dfrac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \qquad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1\ e_2 : \tau_2}$

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- `try "coucou" with` $x.e_2 \to$ `"coucou"`
- `try (raise` s`) with` $x.e_2 \to e_2\{s/x\}$
- `raise "Segmentation fault"` signale une exception
- `(fun (`$x{:}\tau$`) ⇒` x`) raise "Segmentation fault"`

fct° prend argmt x de type T et renvoie x
applique fct° à expr raise

Système de types

Str $\dfrac{}{\Gamma \vdash s : \mathrm{str}}$ Var $\dfrac{}{\Gamma, x:\tau \vdash x:\tau}$ Conc $\dfrac{\Gamma \vdash e_1 : \mathrm{str} \qquad \Gamma \vdash e_2 : \mathrm{str}}{\Gamma \vdash e_1 \hat{} e_2 : \mathrm{str}}$

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- A-t-on unicité du type résultat ?

Système de types

Str $\dfrac{}{\Gamma \vdash s : \mathrm{str}}$ Var $\dfrac{}{\Gamma, x:\tau \vdash x:\tau}$ Conc $\dfrac{\Gamma \vdash e_1 : \mathrm{str} \qquad \Gamma \vdash e_2 : \mathrm{str}}{\Gamma \vdash e_1 \hat{} e_2 : \mathrm{str}}$

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- A-t-on unicité du type résultat ? Non
- $\vdash$ `fun (`x`:str) ⇒ raise "coucou" : str → str`

fct° qui déclenche except° «coucou»
Type spécifié comme une fct° qui prend et renvoie chaines de caracteres

Système de types

Str $\dfrac{}{\Gamma \vdash s : \mathrm{str}}$ Var $\dfrac{}{\Gamma, x:\tau \vdash x:\tau}$ Conc $\dfrac{\Gamma \vdash e_1 : \mathrm{str} \qquad \Gamma \vdash e_2 : \mathrm{str}}{\Gamma \vdash e_1 \hat{} e_2 : \mathrm{str}}$

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Try $\dfrac{\Gamma \vdash e_1 : \tau \qquad \Gamma, x:\mathrm{str} \vdash e_2 : \tau}{\Gamma \vdash \texttt{try}\ e_1\ \texttt{with}\ x.e_2 : \tau}$ Raise $\dfrac{\Gamma \vdash e : \mathrm{str}}{\Gamma \vdash \texttt{raise}\ e : \tau}$

- A-t-on unicité du type résultat ? Non
- $\vdash$ `fun (`x`:str) ⇒ raise "coucou" : str → str` Oui
- $\vdash$ `raise (fun (`x`:str) ⇒` x`) : ?`

Système de types

Str
$$\frac{}{\Gamma \vdash s : \mathsf{str}}$$

Var
$$\frac{}{\Gamma, x : \tau \vdash x : \tau}$$

Conc
$$\frac{\Gamma \vdash e_1 : \mathsf{str} \qquad \Gamma \vdash e_2 : \mathsf{str}}{\Gamma \vdash e_1 \hat{} e_2 : \mathsf{str}}$$

Fun
$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \mathtt{fun}\ (x{:}\tau_1)\ \Rightarrow\ e : \tau_1 \to \tau_2}$$

App
$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \qquad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1\ e_2 : \tau_2}$$

Try
$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma, x : \mathsf{str} \vdash e_2 : \tau}{\Gamma \vdash \mathtt{try}\ e_1\ \mathtt{with}\ x.e_2 : \tau}$$

Raise
$$\frac{\Gamma \vdash e : \mathsf{str}}{\Gamma \vdash \mathtt{raise}\ e : \tau}$$

- A-t-on unicité du type résultat ? Non
- $\vdash \mathtt{fun}\ (x{:}\mathsf{str})\ \Rightarrow\ \mathtt{raise}\ \text{"coucou"} : \mathsf{str} \to \mathsf{str}$ Oui
- $\vdash \mathtt{raise}\ (\mathtt{fun}\ (x{:}\mathsf{str})\ \Rightarrow\ x) : ?$ Pas typable
- $\vdash \mathtt{raise}\ (\mathtt{raise}\ \text{"coucou"}) : \mathsf{str}$

Système de types

Str
$$\frac{}{\Gamma \vdash s : \mathsf{str}}$$

Var
$$\frac{}{\Gamma, x : \tau \vdash x : \tau}$$

Conc
$$\frac{\Gamma \vdash e_1 : \mathsf{str} \qquad \Gamma \vdash e_2 : \mathsf{str}}{\Gamma \vdash e_1 \hat{} e_2 : \mathsf{str}}$$

Fun
$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \mathtt{fun}\ (x{:}\tau_1)\ \Rightarrow\ e : \tau_1 \to \tau_2}$$

App
$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \qquad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1\ e_2 : \tau_2}$$

Try
$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma, x : \mathsf{str} \vdash e_2 : \tau}{\Gamma \vdash \mathtt{try}\ e_1\ \mathtt{with}\ x.e_2 : \tau}$$

Raise
$$\frac{\Gamma \vdash e : \mathsf{str}}{\Gamma \vdash \mathtt{raise}\ e : \tau}$$

- A-t-on unicité du type résultat ? Non
- $\vdash \mathtt{fun}\ (x{:}\mathsf{str})\ \Rightarrow\ \mathtt{raise}\ \text{"coucou"} : \mathsf{str} \to \mathsf{str}$ Oui
- $\vdash \mathtt{raise}\ (\mathtt{fun}\ (x{:}\mathsf{str})\ \Rightarrow\ x) : ?$ Pas typable
- $\vdash \mathtt{raise}\ (\mathtt{raise}\ \text{"coucou"}) : \mathsf{str}$ Oui

6 / 9

Réduction

AppD
$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'}$$

AppG
$$\frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v}$$

Beta
$$\frac{}{(\mathtt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

Réduction

AppD
$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'}$$

AppG
$$\frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v}$$

Beta
$$\frac{}{(\mathtt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

7 / 9

Réduction

AppD
$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'}$$

AppG
$$\frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v}$$

Beta
$$\frac{}{(\mathtt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

Try
$$\frac{e_1 \to e_1'}{\mathtt{try}\ e_1\ \mathtt{with}\ x.e_2 \to \mathtt{try}\ e_1'\ \mathtt{with}\ x.e_2}$$

TryV
$$\frac{}{\mathtt{try}\ v\ \mathtt{with}\ x.e_2 \to v}$$

Réduction

AppD
$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'}$$

AppG
$$\frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v}$$

Beta
$$\frac{}{(\mathtt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

Try
$$\frac{e_1 \to e_1'}{\mathtt{try}\ e_1\ \mathtt{with}\ x.e_2 \to \mathtt{try}\ e_1'\ \mathtt{with}\ x.e_2}$$

TryV
$$\frac{}{\mathtt{try}\ v\ \mathtt{with}\ x.e_2 \to v}$$

TryR
$$\frac{}{\mathtt{try}\ (\mathtt{raise}\ s)\ \mathtt{with}\ x.e_2 \to e_2\{s/x\}}$$

Réduction

AppD
$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'}$$

AppG
$$\frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v}$$

Beta
$$\frac{}{(\texttt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

Try
$$\frac{e_1 \to e_1'}{\texttt{try}\ e_1\ \texttt{with}\ x.e_2 \to \texttt{try}\ e_1'\ \texttt{with}\ x.e_2}$$

TryV
$$\frac{}{\texttt{try}\ v\ \texttt{with}\ x.e_2 \to v}$$

TryR
$$\frac{}{\texttt{try}\ (\texttt{raise}\ s)\ \texttt{with}\ x.e_2 \to e_2\{s/x\}}$$

Raise
$$\frac{e \to e'}{\texttt{raise}\ e \to \texttt{raise}\ e'}$$

Réduction

AppD
$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'}$$

AppG
$$\frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v}$$

Beta
$$\frac{}{(\texttt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

Try
$$\frac{e_1 \to e_1'}{\texttt{try}\ e_1\ \texttt{with}\ x.e_2 \to \texttt{try}\ e_1'\ \texttt{with}\ x.e_2}$$

TryV
$$\frac{}{\texttt{try}\ v\ \texttt{with}\ x.e_2 \to v}$$

TryR
$$\frac{}{\texttt{try}\ (\texttt{raise}\ s)\ \texttt{with}\ x.e_2 \to e_2\{s/x\}}$$

Raise
$$\frac{e \to e'}{\texttt{raise}\ e \to \texttt{raise}\ e'}$$

RaiseAppD
$$\frac{}{e\ \texttt{raise}\ s \to \texttt{raise}\ s}$$

RaiseAppG
$$\frac{}{(\texttt{raise}\ s)\ v \to \texttt{raise}\ s}$$

Réduction

AppD
$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'}$$

AppG
$$\frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v}$$

Beta
$$\frac{}{(\texttt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

Try
$$\frac{e_1 \to e_1'}{\texttt{try}\ e_1\ \texttt{with}\ x.e_2 \to \texttt{try}\ e_1'\ \texttt{with}\ x.e_2}$$

TryV
$$\frac{}{\texttt{try}\ v\ \texttt{with}\ x.e_2 \to v}$$

TryR
$$\frac{}{\texttt{try}\ (\texttt{raise}\ s)\ \texttt{with}\ x.e_2 \to e_2\{s/x\}}$$

Raise
$$\frac{e \to e'}{\texttt{raise}\ e \to \texttt{raise}\ e'}$$

RaiseAppD
$$\frac{}{e\ \texttt{raise}\ s \to \texttt{raise}\ s}$$

RaiseAppG
$$\frac{}{(\texttt{raise}\ s)\ v \to \texttt{raise}\ s}$$

RaiseRaise
$$\frac{}{\texttt{raise}\ (\texttt{raise}\ s) \to \texttt{raise}\ s}$$

Réduction

$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'} \qquad \frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v} \qquad \frac{}{(\texttt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

$$\frac{e_1 \to e_1'}{\texttt{try}\ e_1\ \texttt{with}\ x.e_2 \to \texttt{try}\ e_1'\ \texttt{with}\ x.e_2} \qquad \frac{e \to e'}{\texttt{raise}\ e \to \texttt{raise}\ e'}$$

$$\frac{}{\texttt{try}\ (\texttt{raise}\ s)\ \texttt{with}\ x.e_2 \to e_2\{s/x\}} \qquad \frac{}{\texttt{try}\ v\ \texttt{with}\ x.e_2 \to v}$$

$$\frac{}{e\ \texttt{raise}\ s \to \texttt{raise}\ s} \qquad \frac{}{(\texttt{raise}\ s)\ v \to \texttt{raise}\ s}$$

$$\frac{}{\texttt{raise}\ (\texttt{raise}\ s) \to \texttt{raise}\ s}$$

- $(\texttt{fun}\ (x{:}\texttt{str})\ \Rightarrow\ x)\ \texttt{raise}\ \texttt{"coucou"} \to$?

Réduction

$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'} \qquad \frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v} \qquad \frac{}{(\texttt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

$$\frac{e_1 \to e_1'}{\texttt{try}\ e_1\ \texttt{with}\ x.e_2 \to \texttt{try}\ e_1'\ \texttt{with}\ x.e_2} \qquad \frac{e \to e'}{\texttt{raise}\ e \to \texttt{raise}\ e'}$$

$$\frac{}{\texttt{try}\ (\texttt{raise}\ s)\ \texttt{with}\ x.e_2 \to e_2\{s/x\}} \qquad \frac{}{\texttt{try}\ v\ \texttt{with}\ x.e_2 \to v}$$

$$\frac{}{e\ \texttt{raise}\ s \to \texttt{raise}\ s} \qquad \frac{}{(\texttt{raise}\ s)\ v \to \texttt{raise}\ s}$$

$$\frac{}{\texttt{raise}\ (\texttt{raise}\ s) \to \texttt{raise}\ s}$$

- $(\texttt{fun}\ (x{:}\texttt{str})\ \Rightarrow\ x)\ \texttt{raise}\ \texttt{"coucou"} \to$?
- $\texttt{try fun}\ (x{:}\texttt{str})\ \Rightarrow\ \texttt{raise}\ \texttt{"coucou"}\ \texttt{with}\ y.y\ \texttt{\^{}}\ \texttt{"salut"} \to$?

Réduction

$$\frac{e_2 \to e_2'}{e_1\ e_2 \to e_1\ e_2'} \qquad \frac{e_1 \to e_1'}{e_1\ v \to e_1'\ v} \qquad \frac{}{(\texttt{fun}\ (x{:}\tau)\ \Rightarrow\ e)\ v \to e\{v/x\}}$$

$$\frac{e_1 \to e_1'}{\texttt{try}\ e_1\ \texttt{with}\ x.e_2 \to \texttt{try}\ e_1'\ \texttt{with}\ x.e_2} \qquad \frac{e \to e'}{\texttt{raise}\ e \to \texttt{raise}\ e'}$$

$$\frac{}{\texttt{try}\ (\texttt{raise}\ s)\ \texttt{with}\ x.e_2 \to e_2\{s/x\}} \qquad \frac{}{\texttt{try}\ v\ \texttt{with}\ x.e_2 \to v}$$

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- Si $e \to e'$, alors $(\texttt{raise}\ s)\ e \to$?

Réduction

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$$\frac{e_1 \to e_1'}{\texttt{try}\ e_1\ \texttt{with}\ x.e_2 \to \texttt{try}\ e_1'\ \texttt{with}\ x.e_2} \qquad \frac{e \to e'}{\texttt{raise}\ e \to \texttt{raise}\ e'}$$

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- $(\texttt{fun}\ (x{:}\texttt{str})\ \Rightarrow\ x)\ \texttt{raise}\ \texttt{"coucou"} \to \texttt{raise}\ \texttt{"coucou"}$
- $\texttt{try}\ \texttt{fun}\ (x{:}\texttt{str})\ \Rightarrow\ \texttt{raise}\ \texttt{"coucou"}\ \texttt{with}\ y.y\ \texttt{\^{}}\ \texttt{"salut"} \to \texttt{fun}\ (x{:}\texttt{str})\ \Rightarrow\ \texttt{raise}\ \texttt{"coucou"}$
- Si $e \to e'$, alors $(\texttt{raise}\ s)\ e \to (\texttt{raise}\ s)\ e'$

Sûreté du typage

Théorème (Progression)

Pour toute expression close e et type τ, si $\vdash e : \tau$, alors

Sûreté du typage

Théorème (Progression)

Pour toute expression close e et type τ, si $\vdash e : \tau$, alors

- *e est une valeur, ou*
- *e est une erreur $\texttt{raise}\ s$, ou*
- *il existe e' tel que $e \to e'$.*

Théorème (Préservation)

Pour toutes expressions closes e, e et type τ, si $\vdash e : \tau$ et $e \to e'$, alors $\vdash e' : \tau$.